

YAP and TAZ: a signalling hub of the tumour microenvironment

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Abstract | YAP and TAZ are transcriptional activators pervasively induced in several human solid tumours and their functions in cancer cells are the focus of intense investigation. These studies established that YAP and TAZ are essential to trigger numerous cell-autonomous responses, such as sustained proliferation, cell plasticity, therapy resistance and metastasis. Yet tumours are complex entities, wherein cancer cells are just one of the components of a composite “tumour tissue”. The other component, the tumour stroma, is composed of an extracellular matrix with aberrant mechanical properties and other cell types, including cancer-associated fibroblasts and immune cells. The stroma entertains multiple and bidirectional interactions with tumour cells, establishing dependencies essential to unleash tumorigenesis. The molecular players of such interplay remain partially understood. Here, we review the emerging role of YAP and TAZ in choreographing tumour–stromal interactions. YAP and TAZ act within tumour cells to orchestrate responses in stromal cells. Vice versa, YAP and TAZ in stromal cells trigger effects that positively feed back on the growth of tumour cells. Recognizing YAP and TAZ as a hub of the network of signals exchanged within the tumour microenvironment provides a fresh perspective on the molecular principles of tumour self-organization, promising to unveil numerous new vulnerabilities.

Intratumoural phenotypic heterogeneity

The coexistence, within the same tumour, of cellular populations that differ in their molecular and biological properties, including clinically important phenotypes such as the ability to seed metastases and to survive therapy.

Solid tumours are complex entities, in which transformed tumour cells are surrounded by a heterogeneous array of non-transformed stromal cells and extracellular matrix (ECM)^{1,2}. This tissue organization represents a recapitulation of cell–cell and cell–ECM interplays that orchestrate organ development and wound healing, as famously epitomized by Harold Dvorak's statement that tumours are “wounds that do not heal”³. Remarkably, the influence of the stroma on tumour initiation and progression is overarching: on the one hand, the microenvironment of normal tissues is able to reduce, if not “normalize”, the effects of aggressive oncogenes in epithelial cells^{4,5}, to the extent that normal cells of healthy tissues carry a host of oncogenic alterations without evolving into tumours^{6,7}. On the other hand, once tumour cells acquire the ability to escape such normalizing cues, they still require a corrupted, pro-tumorigenic version of the microenvironment to boost their progression and metastasis⁴. It is possible that normalizing and tumorigenic ecosystems coexist and affect genetically identical cells of the same tumour, contributing to intratumoural phenotypic heterogeneity. Tumour cells never become entirely independent from the tumour microenvironment (TME), which bears important translational potential: through targeted therapy of genetically stable stromal cells, the TME can be rehabilitated into a

tumour-suppressive microenvironment with reduced risks of resistance⁵.

Oncogenic lesions and stromal alterations cooperate in the poorly understood mechanisms of tumour morphogenesis, by inducing changes in cell shape and cell fate. The latter empower cancer cells with several attributes including increased stemness and migratory and proliferative potentials⁸, but also the ability to orchestrate new microenvironmental changes. In turn, such changes induce a new wave of dynamic negotiations and mutual adaptations between tumour cells and their stroma, which is similar to a developmental process.

Here, we review how the function and regulation of YAP and TAZ^{9–11} stand at the centrepiece of a signalling hub, shared by epithelial tumour cells and stromal cells alike, which is critical for the build-up of tumour tissue. YAP and TAZ capture information from the physical context that the cell experiences (BOX 1), and turn it into a transcriptional response^{12–14} (BOX 2). YAP and TAZ are involved in development and tissue regeneration, and their aberrant activation is a hallmark of several malignancies¹² (BOX 3). First, we will outline the attributes conferred by YAP and TAZ to tumour cells (FIG. 1a). Then, we will move to a “tissue perspective” on YAP and TAZ biology in cancer, expanding aspects that have so far received little attention but emerge as

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Box 1 | Overview of mechanisms regulating YAP and TAZ activity

YAP and TAZ are two highly related transcriptional regulators whose activation level is the result of the integration of various signals, briefly summarized and reviewed in detail by recent contributions^{12,13,125}.

- YAP and TAZ sense and respond to the physical cues that each cell receives from the surrounding microenvironment and the whole tissue. Mechanoregulation of YAP and TAZ depends on the structural organization and tension of the F-actin cytoskeleton, which receives instructive stimuli from cell–cell adhesion and cell–extracellular matrix attachment through integrins, focal adhesions and other mechanosensors also involving Rho GTPases (reviewed elsewhere^{73,126}).
- YAP and TAZ are negatively regulated by the kinases of the Hippo pathway. Polarity complexes in epithelial cells and the tumour suppressor neurofibromin 2 (NF2, also known as Merlin) are important activators of the Hippo kinases, and thus inhibitors of YAP and TAZ transcriptional activity¹²⁷.
- YAP and TAZ respond to growth factors and soluble signals through crosstalk with many signalling pathways, including WNT, transforming growth factor- β , Hedgehog and epidermal growth factor receptor signalling. Importantly, YAP and TAZ themselves contribute to the regulation of several of these pathways, in a reciprocal cross-regulation^{13,125}. Moreover, YAP and TAZ entertain crosstalk with Notch signalling by controlling the expression of Delta ligands¹²⁸.
- Activation of various G-protein-coupled receptors by their respective ligands also impacts on YAP and TAZ activity, in an activating or inhibitory manner depending on the G-protein α -subunit involved¹²⁷.
- The cellular metabolic state has a profound effect on YAP and TAZ activity; generally speaking, nutrient availability (glucose, lipids, amino acids) favours YAP and TAZ activity³³.

pivotal for tumour growth and metastasis. Acting within tumour cells, YAP and TAZ substantially contribute to orchestrating the responses of stromal cells (FIG. 1b); vice versa, YAP and TAZ activation in stromal cells triggers effects that positively feed back on the growth of tumour cells, creating mutually reinforcing vicious cycles and the progressive build-up of corrupted TMEs. Thus, YAP and TAZ activity within different cells sculpts the organization of the whole tumour tissue. Ultimately, we aim to convey the message that YAP and TAZ are precious tools to gain unprecedented insights into how tumour–stromal interactions are orchestrated and into the molecular principles of tumour self-organization.

YAP and TAZ and the origin of human cancer

YAP and TAZ are transcriptional activators that are pervasively induced in most, if not all, human solid tumours, as revealed by immunohistochemistry and/or gene-expression analyses in large cohorts of patients^{15,16}. Functionally, YAP and TAZ are instrumental for tumour initiation and/or progression in multiple tissue types, as shown in several experimental model systems¹⁵ (BOX 3). This role of YAP and TAZ in tumour growth parallels its essentiality for tissue regeneration but is in stark contrast with the dispensability of YAP and TAZ for the normal homeostasis of the respective tissues¹⁷. A scenario to explain this conundrum could be that, in physiological conditions, YAP and TAZ act as “master genes” of tissue repair, activated by stressful conditions that disrupt normal tissue architecture to induce regrowth of the damaged or missing tissues from the remaining cells. This regenerative potential is used over and over by tumour cells, and favours their stable escape from tumour-suppressive tissue-level checkpoints.

The contrast between YAP and TAZ dispensability in stem cells for normal homeostasis versus essentiality

for tissue repair and tumorigenesis also raises still unanswered questions on the identity of the cancer cell of origin: is cancer in fact the product not of a professional stem cell but of a more differentiated cell, converted into a stem-like cell upon YAP and TAZ activation (see Phenotypic plasticity)? Several lines of evidence indicate that this is the case: for example, breast cancer emerges invariably from luminal cells¹⁸, lung cancer from alveolar type II cells¹⁹, glioblastoma from astrocyte-like cells²⁰, pancreatic cancer from pancreatic acinar cells²¹, liver cancer from hepatocytes and bile duct cells²², and so forth. In all of these normal cell types, YAP and TAZ are typically undetectable, likely explaining their dispensability in homeostatic conditions, but become active in their tumour cell derivatives¹⁵.

Cell-autonomous functions of YAP and TAZ

YAP and TAZ activation endows epithelial cells with several of the “hallmarks of cancer”, and in particular proliferation, phenotypic plasticity, drug resistance and metastasis¹⁵. Importantly, several of these attributes can be directly induced by transient expression of YAP and TAZ in normal primary cells purified from healthy tissues²³, suggesting that YAP and TAZ can extensively rewire the epigenome of even terminally differentiated cells, reprogramming them into stem-like cells. Once YAP and TAZ become constitutively active, for example in advanced tumours, YAP and TAZ-expressing cells start displaying increased malignant potential as outlined below (FIG. 1a).

Cell proliferation. YAP and TAZ stimulate cell proliferation by controlling the expression of a broad number of cell cycle regulators, of factors involved in DNA duplication and DNA repair, and of factors involved in mitosis^{24–26}. In addition, YAP and TAZ amplify their own effects, by activating self-sustaining positive loops through their target genes^{27,28} and by turning on the expression and potentiating the transcriptional responses of other proto-oncogenes, for example MYC and AP-1 family members (JUN and FOS-like factors)^{24,29–32} (see also BOX 2). YAP and TAZ also support the high metabolic requirements of proliferating cancer cells by transcriptionally upregulating enzymes involved in glycolysis, glutamine metabolism and nucleotide biosynthesis^{33–36}.

Phenotypic plasticity. Overexpression of exogenous YAP and TAZ in primary normal differentiated cells converts them into stem-like cells of the same tissue. For example, introducing YAP and TAZ into luminal differentiated cells of the mammary gland converts them into mammary stem cells, able to regenerate a secondary mammary tree once implanted in vivo²³. This striking cell-fate reprogramming potential also operates in other cellular contexts, such as pancreatic acinar cells, hepatocytes, neurons and astrocytes^{23,37}, all cells that correspond to the cell of origin of the tumours emerging in those tissues. This indicates that YAP and TAZ are shared inducers of various “stemness networks” in different histotypes. The reprogramming trajectories induced by YAP and TAZ to build new stem cells from normal differentiated cells remain unknown, but it is tempting to speculate that these trajectories recapitulate

Self-organization

An emergent property of normal and tumour cell ensembles, by which patterns of differentiation and ordered tissue architectures occur as a product of cell–cell and cell–extracellular matrix contacts.

Cell of origin

The normal cell that is hit by the first transforming events, eventually growing into a tumour; depending on the tumour type, it can be either a differentiated cell or a resident stem cell.

Epigenome

A map of the genome-wide modifications made to DNA and the proteins associated with DNA that influence gene availability for transcription.

Self-sustaining positive loops

A circuit whereby a factor stabilizes or enhances its own activation by promoting the expression or function of a positive regulator.

Cell-fate reprogramming

A non-physiological change in cell identity, either to a less differentiated status or to a completely different lineage.

Cancer stem cells

The tumour cells that maintain tumour propagation, by virtue of their self-renewal capacity and multipotency.

Epithelial-to-mesenchymal transition

The acquisition, by epithelial cells, of a mesenchymal gene programme, which facilitates migration and invasion.

Chemotherapeutic drugs

Standard anticancer drugs that target cell division and DNA replication.

Molecularly targeted drugs

Drugs that target specific molecular alterations (such as mutated oncogenes) in cancer cells.

Cytoskeletal remodelling

Dynamic reshaping of the actin cytoskeleton, which is essential for the cell to adapt to changes in the physical properties of the extracellular environment or to migrate.

developmental processes. For example, YAP and TAZ are essential for intestinal recovery after injury by converting the remaining cells of the repairing epithelium not into adult intestinal stem cells but rather into fetal-like intestinal cells^{38–40}. It remains to be established whether this phenomenon also occurs in other contexts, and whether this effect contributes to the roles of YAP and TAZ as tumour constructors. That said, YAP and TAZ activation causes the transition of immortalized cells into cancer stem cells⁴¹. At least superficially, stemness induced by YAP and TAZ echoes the effect of epithelial-to-mesenchymal transition (EMT)⁴². The connections between YAP, TAZ and EMT or between YAP, TAZ and EMT inducing transcription factors are only partially understood⁴³. YAP overexpression has been reported in some contexts to support EMT^{9,44}; however, other studies suggest that whereas EMT-induced stemness induces and requires endogenous YAP and TAZ, cell plasticity induced by gain of YAP and TAZ does not necessarily require an EMT^{41,45}.

Therapy resistance. Another key feature of cancer cells supported by YAP and TAZ is resistance to antitumour treatments, including chemotherapeutic drugs, radiation and molecularly targeted drugs^{46,47}. In drug-resistant cells, YAP and TAZ are activated as an alternative survival pathway, for example in melanoma, colorectal and thyroid tumour cells treated with RAF and MEK inhibitors

in vitro^{48,49}. These results resonate with in vivo findings in RAS-induced pancreatic tumours in transgenic mice, whereby activation of YAP is causal for tumour recurrence after sudden elimination of the initiating oncogene^{25,50}. YAP and TAZ can also participate in the molecular mechanisms that induce drug resistance, for example in cells treated with CDK4/6 inhibitors, where YAP and TAZ upregulate the expression of CDK6 to overcome its pharmacological inhibition⁵¹. Thus, inhibition of YAP and TAZ may be a valuable tool to prevent the emergence of drug resistance or to restore drug sensitivity^{48,49,52,53}.

Metastasis. There is increasing recognition that several steps of the invasion–metastasis cascade are induced by and require YAP and TAZ (FIG. 2). This has been recently reviewed⁵⁴, but further recent evidence offers additional insights. The cytoskeleton needs to be continually reorganized as cancer cells change shape and move, to invade the surrounding tissue, extravasate and reach the metastatic site. YAP and TAZ not only sense cytoskeletal tension but also regulate F-actin dynamics, by promoting the transcription of genes encoding Rho-GTPase activating proteins (for example, ARHGAP29); in so doing, YAP and TAZ favour cytoskeletal remodelling in cancer cells⁵⁵. YAP and TAZ also maintain plasticity in cell–ECM adhesion⁵⁶ by limiting the maturation of focal adhesions, enabling cell migration overall. Once tumour cells have abandoned the primary tumour and entered the blood or lymphatic circulation, YAP and TAZ are responsible for resistance to anoikis⁵⁷. In addition, platelets have been reported to promote YAP activation through an as yet unidentified signal⁵⁸, explaining, at least in part, how the contact between circulating tumour cells and platelets favours cancer cell survival and metastatic spreading. Metastatic cells also require specific metabolic adaptations: lymph-node metastasis involves a shift towards fatty acid oxidation, which is mediated by YAP and TAZ; interestingly, fatty acid oxidation inhibitors recapitulate YAP and TAZ inhibition, suppressing lymph-node metastasis⁵⁹.

Non-cell-autonomous functions of YAP and TAZ

YAP and TAZ coordinate the behaviour of cells within the complex tumour ecosystem, partly through the short-range communication between newly-transformed cells and neighbouring normal cells, as outlined below (FIG. 3).

Cell competition. Healthy tissues display tumour-preventing properties that eliminate abnormal, transformed and even cancerous cells to maintain the normal epithelium (FIG. 3a). The competitive nature of such tumour–host interaction, however, is entirely bidirectional; tumour cells can eventually induce death of the surrounding healthy cells, as such “making space” for their own expansion^{60,61}. Clearly, understanding how to enhance the tumour-suppressive properties of the host tissue might offer unexpected insights into preventing tumorigenesis and recurrence. Cell competition has mainly been studied in *Drosophila melanogaster*. However, recent studies have indicated that cell competition also operates in mammalian tissues and that YAP

Box 2 | Mechanisms of YAP and TAZ-dependent transcription

In the nucleus, YAP and TAZ need to interact with a DNA-binding transcription factor to activate transcription, as they lack a DNA-binding domain. The main (but not unique) partners of YAP and TAZ are TEAD family members^{13,125}; interestingly, TEAD association with DNA is quite long-lived compared with the highly dynamic interaction between YAP and TEAD¹²⁹, suggesting that TEAD might be “pre-loaded” on chromatin and YAP and TAZ might then deliver an activation signal. The complex of YAP and TAZ with TEAD mainly binds distal enhancer elements^{24,79,130}, which contact target promoters through chromatin looping. In adult mouse cardiomyocytes, overexpressed YAP can establish new chromatin loops¹³¹; it is still unclear whether YAP and TAZ can modify the genome architecture also in cancer cells. In any case, YAP and TAZ are essential for the establishment of a histone code that promotes the recruitment of RNA polymerase II to the promoters of their target genes; YAP and TAZ do so by physically recruiting the histone acetyltransferase p300 (REF. 79) and BRD4 (REF. 53). Beyond transcriptional initiation, YAP and TAZ also promote efficient transcript elongation by promoting the release of RNA polymerase II from promoter proximal pausing¹³⁰. YAP and TAZ epigenetic studies also revealed that these transcriptional regulators are often flanked by activator protein 1 (AP-1) transcription factors, recruited at AP-1 binding sites located side by side with TEAD-binding elements; AP-1, YAP or TAZ and TEAD transcriptionally cooperate in several contexts, jointly regulating tumour cell proliferation and motility^{24,27,29,30,79}.

Cancer cells in which YAP and TAZ are activated become dependent on YAP and TAZ themselves, and on the sustained expression of their pro-tumorigenic target genes. In particular, the connection between YAP and TAZ and BRD4 is a main culprit of the “transcriptional addiction” phenomenon in solid tumours, whereby cancer cells establish transcriptional dependencies that may be amenable to therapeutic targeting. Indeed, treatment with BRD4-inhibitory drugs (known as BET inhibitors, currently under clinical evaluation) is sufficient to oppose YAP and TAZ pro-tumorigenic functions in vitro and in vivo^{32,53}.

Recently, some nuclear inhibitors of YAP and TAZ have been described, including VGLL4, the SWI/SNF subunit ARID1a, nuclear TIAM1 and the long non-coding RNA MALAT1. All of these molecules compete with YAP and TAZ for TEAD binding, resulting in the downregulation of YAP and TAZ target genes^{45,132–134}. Mimicking these molecules, or favouring their interaction with TEAD over YAP–TEAD association, might be a strategy to inhibit YAP and TAZ¹³⁵.

Box 3 | YAP and TAZ in tumours: evidence from clinical samples and animal models

Breast cancer

YAP and TAZ activation correlates with grade, metastasis proclivity and failure to respond to therapies in HER2-positive breast cancer and triple-negative breast cancer (TNBC) (reviewed elsewhere¹⁵). Functionally, in mouse models, TAZ activity is sufficient and required for tumorigenesis and metastasis from transplanted immortalized human cells^{41,136}, and conditional mammary gland knockout of YAP in spontaneous tumour models (that is, MMTV-PyMT) extends tumour latency¹³⁷. Strikingly, combined genetic loss of YAP and TAZ completely abolishes initiation of tumorigenesis in another, WNT-driven, TNBC-like mammary tumour model (that is, after conditional APC loss)⁵³.

Lung cancer

Nuclear YAP and TAZ levels and their signatures correlate with grading and metastasis in non-small-cell lung cancer (NSCLC) patients (reviewed elsewhere¹³⁸). In NSCLC mouse models, YAP is critical for tumour progression and metastasis of tumours induced by combined gain of Ras (*Ras*^{G12D}) and loss of the polarity master gene *Lkb1* (REF.¹³⁹).

Liver cancer

Hepatocellular carcinomas (HCCs) and cholangiocarcinomas (CCs) display elevated YAP and TAZ activity levels, correlated with poor outcome¹⁴⁰. In mouse models, YAP and TAZ are essential for HCC and CC⁴⁵. Intriguingly, YAP and TAZ are required in hepatocytes to allow CC development⁴⁵, suggesting that YAP and TAZ promote the secretion of as yet unknown tumour-promoting signals, or that YAP and TAZ can mediate the return to an early, bipotent liver progenitor state³⁷.

Pancreatic cancer

YAP and TAZ nuclear staining becomes detectable in benign lesions and strong in pancreatic ductal adenocarcinomas¹⁴¹. Genetically, YAP is essential for malignant transition⁸⁷, although the potential for TAZ to compensate at earlier stages for tumour initiation remains unknown.

Skin cancer

YAP and TAZ are activated in human squamous cell carcinomas and basal cell carcinomas, and they are required for tumour formation in mouse models of these tumour types^{29,142}.

and TAZ are central, evolutionarily conserved players in these phenomena^{62,63}. Loss of the polarity factor Scribble, which frequently occurs in tumours⁶⁴, can disable the Hippo pathway leading to activation of Yorkie (the *Drosophila* homologue of YAP)⁶⁵. However, when an individual Scribble-deficient cell is surrounded by normal cells, it undergoes compaction and is eventually evicted from the monolayer⁶⁶ (FIG. 3b), indicating that the tissue context dominates over inactivation of the Hippo pathway. Of note, cell compaction is an important mechanical inhibitory input of YAP and TAZ⁶⁷, potentially overcoming the effect of Scribble inactivation and preventing YAP and TAZ activation. By contrast, when Scribble-mutant fly tumour cells are allowed to accumulate (for example, by concomitant expression of oncogenic *Ras*) they become supercompetitors, because unabated Yorkie activation increases their fitness, thereby inducing death of nearby normal cells⁶² (FIG. 3c). Of note, it is not the absolute but the relative level of Yorkie that defines the loser versus winner competitive cell state. Gliomas originally display heterogeneous expression of YAP in different cell clones; however, tumour progression leads to clonal dominance of those cells expressing the highest YAP level, as these induce apoptosis of cells expressing the lesser amounts of YAP⁶⁸. Intestinal tumorigenesis is well known to originate from gain of WNT signalling after genetic loss of the APC tumour suppressor. YAP and TAZ are downstream of WNT signalling, potentially stabilized in colorectal cancer and required for intestinal tumorigenesis^{69–71}. This requirement is evolutionarily conserved in *Drosophila*,

where Yorkie fuels the outgrowth of *Apc*-mutant intestinal tumours, in part through cell competition. However, if Yorkie levels are experimentally sustained in *Apc*-wild-type neighbouring cells (by partial (heterozygous) inactivation of the Hippo pathway), these normal cells flip to a “supercompetitor” state and tumour growth is confined⁷² (FIG. 3d).

Sharing the ECM. YAP and TAZ are universal readers of cell shape and of the mechanical inputs the cells receive from their location within the 3D tissue architecture, that is, from neighbouring cells and their attachment to the substrate. This is true for epithelial as well as stromal compartments. In particular, the structural and physical attributes of the ECM shared by all cells of the tumour tissue are dominant determinants of YAP and TAZ regulation. YAP and TAZ are inhibited by the physiological “softness” of the ECM that characterizes healthy tissues but induced by supra-physiological ECM rigidity⁷³. We propose that the compliance of normal tissues that keeps YAP and TAZ activity down is a main tumour-suppressor mechanism of our bodies, constraining the effects of oncogenic lesions, and might contribute to the reason why some tumours are rare, typically benign or undetectable for decades^{4,74}. Ageing, inflammation and tissue damage cause ECM stiffening and/or altered mechanical homeostasis, lowering the threshold for YAP and TAZ activation. At the same time, other signalling conduits (see BOX 1) further pattern YAP and TAZ activity. For example, apico-basal cell polarity and cell–cell adhesion through adherens junctions feed on the Hippo kinases, potentially inhibiting YAP and TAZ: these cellular features, dynamically lost and regained by tumour cells as part of their inherent phenotypic plasticity, may cooperate with the effect of a compliant ECM or potentially enhance the effects of a rigid ECM. Thus, the activity status of YAP and TAZ is influenced by extrinsic and intrinsic structural cues of tumour cells and their environment, as such contributing to benign versus malignant tumour cell behaviour.

The tumour ECM, a key component of the TME, is the medium shared by tumour and stromal cells with which they dynamically interact to exchange physical and chemical signals. Cancer-associated fibroblasts (CAFs) are the main cell type responsible for the tumour-specific changes in the ECM, and this includes increased ECM deposition, collagen crosslinking and expression of matrix metalloproteinases². During tumour evolution, the ECM becomes progressively stiffer and aggressive tumours are characterized by a strong desmoplastic reaction at the tumour border, with accumulation of multiple ECM proteins. Laminin, a host of laminin-associated proteins and proteoglycans have also been implicated in mediating tissue stiffness and collagen remodelling that choreograph the tumour landscape^{75,76}. Epithelial tumour cells adhere to such increasingly rigid ECM through integrins and focal adhesions, in turn restructuring the organization and tensional state of their cytoskeleton. This initiates a mechanotransduction cascade that leads to YAP and TAZ activation^{73,77,78}. Tumour cells are not just passive receivers of inputs from the ECM but actively contribute to the physical

Anoikis

Apoptosis induced by loss of cell–cell or cell–extracellular matrix attachment.

Apico-basal cell polarity

A polarized organization of epithelial cells with a specialized apical membrane facing the outside of the body or lumen of internal cavities and a specialized basolateral membrane localized on the opposite side, established and maintained by specific protein complexes.

Desmoplastic reaction

Excessive production of extracellular matrix proteins and extensive proliferation of stromal fibroblasts, resulting in a dense and fibrous connective tissue around tumour cells.

Mechanotransduction

Mechanisms by which cells convert mechanical signals conveyed by the microenvironment into biochemical signals to adapt their behaviour to the environment.

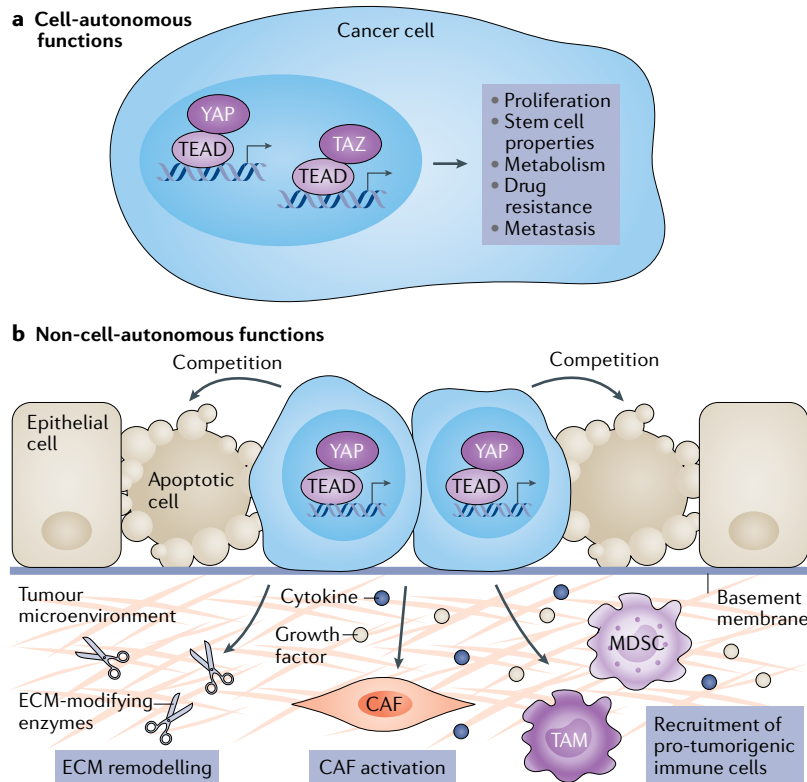


Fig. 1 | Overview of YAP and TAZ functions in cancer cells. **a** | YAP and TAZ activation in cancer cells impacts on the behaviour of cancer cells themselves, by regulating the capacity to proliferate and adjust their metabolism to the altered cellular context, promoting the acquisition of stem-like properties, drug resistance and migratory capacity that allow tissue invasion and metastatic dissemination. **b** | YAP and TAZ activation in cancer cells also impacts on the features of the tumour stroma by modifying the composition and physical properties of the tumour extracellular matrix (ECM) through secretion of ECM components and ECM-modifying enzymes (scissors) and recruitment of pro-tumorigenic cells that populate the tumour stroma, including cancer-associated fibroblasts (CAFs) and pro-tumorigenic immune cells. The latter include tumour-associated macrophages (TAMs) and myeloid-derived suppressor cells (MDSCs) recruited by tumour-derived cytokines. Only YAP is shown in panel **b** for simplicity, but TAZ can carry out the same functions.

ECM-modifying enzymes

Enzymes that modify extracellular matrix (ECM) proteins in the extracellular space; they comprise proteinases that degrade ECM components (for example, matrix metalloproteinases, ADAMTS, plasmin, cathepsins) and enzymes that modify ECM topography (for example, LOX enzymes, which crosslink collagen).

Disseminated cancer cells

Tumour cells that have left the primary tumour, passed through the vascular or lymphatic system and migrated to distant organs.

Dormancy

The condition of a cancer cell that remains alive but proliferatively inert.

properties of ECM fibres, by modifying their composition and fine organization⁷⁶ (FIG. 1b): indeed, YAP target genes transcribed in cancer cells include genes encoding ECM components and ECM-modifying enzymes^{79–81}. Moreover, cells reciprocate the mechanical strains they receive by increasing affinity of their integrin receptors for the ECM⁸². Intriguingly, some YAP and TAZ transcriptional targets are dedicated to preserving such reciprocity, and these include expression of F-actin remodelling factors, focal adhesion docking proteins and $\beta 1$ integrin (REFS^{55,56,83}).

Crucially, stiffening the tumour ECM triggers YAP and TAZ mechanotransduction also within CAFs themselves⁸⁴. YAP activity is an early marker of CAFs and is required for CAF functions, preserving their contractile phenotypes and fostering deposition of thick and rigid collagen fibres that sustain the proliferation of cancer cells (FIG. 4a). Other signals potentiate YAP and TAZ activation in CAFs, including WNT signalling⁸⁵. The symbiotic relationships between CAFs and epithelial cells orchestrated by TME mechanics also extend to

their metabolic communication. Amino acid exchange between CAFs and cancer cells sustain cancer cell proliferation and CAF activation through YAP and TAZ mechanotransduction operating in both cell types⁸⁶. In other words, the abnormal physical attributes of the tumour ECM, through YAP and TAZ mechanotransduction, establish feed-forward, self-reinforcing loops that fuel CAF activity; this is a ratchet-like mechanism, leading to a more corrupted tensional state in cancer cells and, likely, in other cells entering the TME.

The existence of mutually reinforcing loops, whereby YAP and TAZ mechanotransduction serves as a common language between different cells, complicates the dissection of how YAP and TAZ activation is initiated in the TME. Many different and potentially concurrent events may converge to this point: for example, genetic lesions that lead to oncogenic activation of the RAS–PI3K pathway⁸⁷, or the WNT–APC–AXIN pathway⁶⁹ may ignite this system starting from tumour cells; on the other extreme, YAP activation in stromal fibroblasts during inflammation may precede the appearance of tumour cells.

Mechanical crosstalk in metastasis. A critical step in metastasis is the survival of disseminated cancer cells in the alien environment of the colonized organ⁵, as typically warranted by the lodging of extravasated tumour cells within the perivascular niche. Mechanistically, disseminated cancer cells spread on capillaries of the host tissue, behaving similarly to pericytes, and spreading activates YAP and TAZ mechanosignalling, which contributes to tumour cell survival. This mechanism is required both for the growth of aggressive metastatic cells immediately after extravasation and for the awakening from dormancy of disseminated cancer cells⁸⁸. Skewing, or awakening from, such a “dormant” state requires the build-up of a larger and more permissive metastatic niche in the host parenchyma⁵. Once again, at least for dormant breast cancer cells in the lung, YAP and TAZ are critical for such a pivotal step of metastatization, through an unexpected twist: here, an essential role is played by neutrophil-derived extracellular traps, which are scaffolds of DNA with associated proteases released by neutrophils during chronic inflammation, such as after prolonged exposure to cigarette smoke. Neutrophil-derived extracellular trap-associated proteases digest and remodel the ECM, generating cleaved laminin peptides. These peptides activate integrin signalling in dormant cancer cells, which then raises YAP and TAZ activity, cell proliferation and outgrowth of metastatic colonies⁸⁹. Importantly, in humans, smoking is a risk factor for recurrence of breast cancer⁹⁰.

Inflammation. Inflammation by innate immune cells designed to fight infections and heal wounds also supplies the TME with bioactive molecules (for example, IL-6) and ECM-modifying enzymes. Of note, IL-6–gp130 signalling has also been shown to engage YAP activation, to the extent that the consequences of the injury and inflammation cascade have been shown to be entirely YAP-dependent, at least in the intestine⁹¹. It is tempting to speculate that YAP activation by ECM

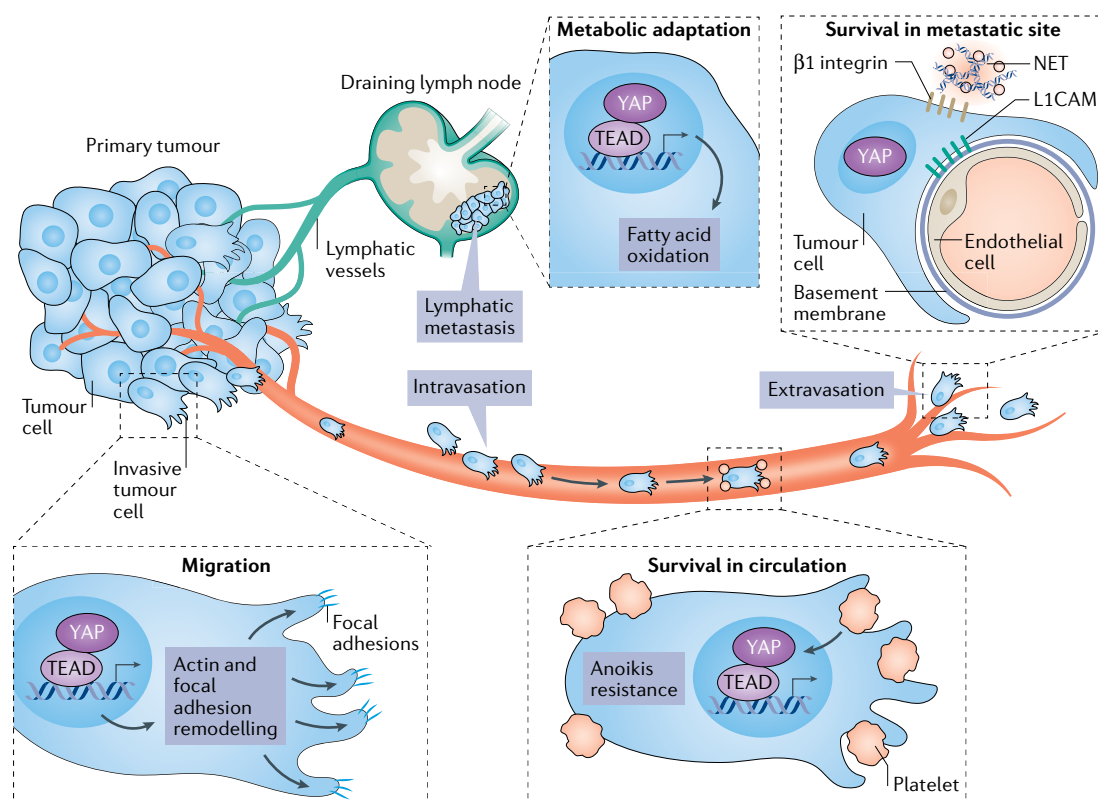


Fig. 2 | New functions of YAP and TAZ in metastasis. YAP and TAZ support the metastatic process in several ways: they maintain the flexibility of the cytoskeleton, enabling migration; they mediate the metabolic adaptation of the cell to the new microenvironment, for example by allowing the switch to fatty acid oxidation in the lipid-rich lymph-node environment; and they respond to signals (such as those coming from platelets) that favour the survival of tumour cells that have left the primary site. In the secondary site, disseminated cancer cells employ cell adhesion molecule L1 (L1CAM) to spread on capillaries in a pericyte-like manner, leading to mechanical activation of YAP and TAZ, which is essential for the survival and growth of metastatic cells⁸⁸. In the metastatic niche, chronic inflammation contributes to mechanical activation of disseminated cells through extracellular matrix remodelling. In part, in the context of cigarette smoke-induced lung inflammation, this is due to the activity of neutrophil-derived extracellular trap (NET)-associated proteases, which can trigger integrin signalling in cancer cells. Only YAP is shown in the figure for simplicity, but TAZ can carry out the same functions.

stiffening and interleukin signalling in the TME may be contributing to the well-known epidemiological association between chronic inflammation or fibrosis with increased risk of malignancy.

Controlling stromal responses

Once YAP and TAZ are active in tumour cells, they are able to modify the cellular composition of their stroma and to reprogramme stromal cell behaviour to contribute to carcinogenesis. Beyond CAFs, prominent examples are immune cells that infiltrate the tumour stroma. Based on several recent studies, a “social” network can be described whereby YAP and TAZ in epithelial cells contribute to orchestrating the immune response to protect incipient tumours from immune surveillance (FIG. 1b).

Tumour-associated macrophages. Within established tumours, macrophages foster tumour development, invasion and metastasis, and their presence is often associated with poor prognosis⁹². Activated macrophages are classified as M1 or M2: M1 macrophages are considered to exert antitumour activity, whereas pro-tumorigenic functions are associated with M2

polarization⁹². There is a wide association between YAP and TAZ activation and macrophage recruitment in several mouse models of liver cancer^{93–96}; YAP overexpression in a single hepatocyte in a healthy mouse liver is sufficient to induce macrophage infiltration⁹⁵. Notably, macrophage depletion impairs tumour development in mouse livers engineered to overactivate YAP and TAZ, suggesting an essential role of macrophages in the tumorigenic process⁹⁶. Whereas macrophage depletion would hardly represent a viable therapeutic option, engaging the mechanisms of YAP and TAZ function in these cells may offer important routes for macrophage re-education from pro-tumorigenic to antitumorigenic. Indeed, YAP-expressing hepatocytes actively recruit macrophages through their direct target gene C–C motif chemokine ligand 2 (*Ccl2*, also known as *Mcp1*), a macrophage-specific chemokine, and silencing of *Ccl2* is indeed sufficient to impair macrophage recruitment and prevent tumour development^{95,96}. Therefore, it will be interesting to directly address whether YAP and TAZ from tumour cells can skew macrophage differentiation towards the M2 phenotype in vivo. It is still to be defined whether YAP and TAZ promote the proliferation and

Neutrophil-derived extracellular traps
Large, extracellular, web-like structures released by neutrophils, composed of cytosolic and granule proteins that are assembled on a scaffold of decondensed chromatin.

Immune surveillance
The ability of the immune system to identify and destroy nascent tumours.

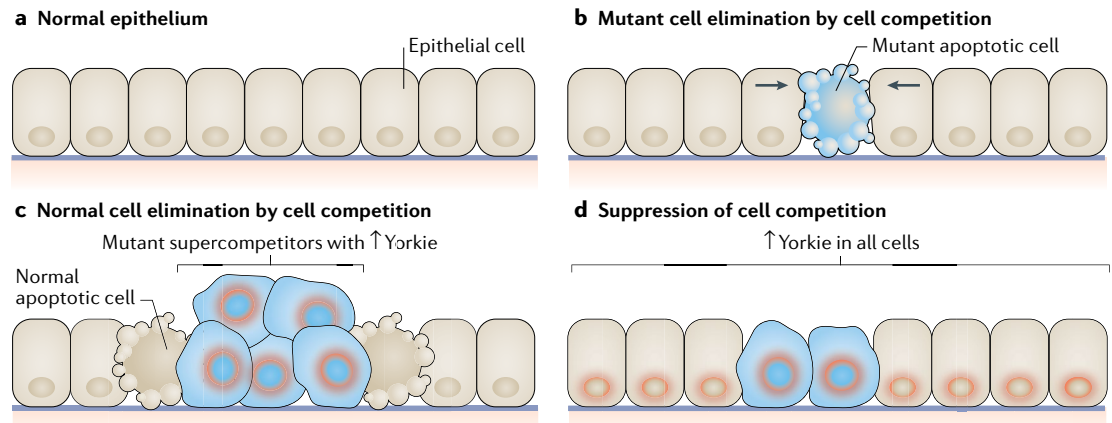


Fig. 3 | Yorkie/YAP and TAZ in the competition between emerging tumour cells and normal epithelial cells. Normal epithelia (part **a**) are tumour-suppressive. A single cell with oncogenic potential (for example, a cell with deranged polarity due to *Scrib* loss of function) is outcompeted by its normal neighbours, which compress it and induce the activation of apoptotic signals (part **b**). However, oncogenic mutations leading to Yorkie/YAP and TAZ accumulation (such as combined *Scrib* and *Ras* mutation, or loss of APC) allow transformed cells to increase their fitness and win the competition with nearby normal cells, allowing expansion of the neoplastic clone (part **c**). In this competition between normal and mutant epithelial cells, the relative level of Yorkie/YAP and TAZ activity determines who wins. In *Drosophila*, raising Yorkie levels (by partial inactivation of the Hippo pathway) in otherwise healthy cells surrounding transformed intestinal cells prevents expansion of the tumorigenic clone (part **d**). In all panels, YAP or TAZ activation in cells is signified by a red glow around the nucleus edge.

activation of the liver-resident macrophages (the Kupffer cells), or whether they can promote the homing of circulating monocytes to the TME, and at which steps of the tumorigenic process YAP-mediated macrophage recruitment becomes essential. Of note, TAZ activity in hepatocytes mediates macrophage accumulation in the inflamed and fibrotic liver parenchyma in mouse models of nonalcoholic steatohepatitis⁹⁷, which often precedes hepatocellular carcinoma. Analysis of the TCGA dataset and immunohistochemistry evaluation of a few human hepatocellular carcinomas suggest that a positive correlation might exist between the expression levels of well-established YAP and TAZ target genes and CCL2 (REF.⁹⁶). This implies that YAP-mediated macrophage recruitment might be relevant for human liver cancer, which is well known to occur after a history of tissue damage, in a context of severe inflammation, cirrhosis and fibrosis. The YAP-dependent tumour cell–macrophage connection might extend to other non-hepatic malignancies; first evidence in this direction exists for YAP-induced macrophage recruitment in a mouse model of pancreatic adenocarcinoma⁹⁸. It is worth noting that macrophages may be mechanically sensitive, and that their phenotype may be regulated, in part, by the same physical cues that control tumour epithelial cells and CAFs within the tumour ECM. For example, when macrophages are forced to assume an elongated cell shape, they upregulate markers of M2 polarization^{99,100}.

Myeloid-derived suppressor cells. Tumour progression is associated with accumulation of myeloid-derived suppressor cells (MDSCs), which establish an immune-suppressive and thus tumour-tolerant milieu by inhibiting cytotoxic T cells; moreover, MDSCs promote tumour progression by modifying the TME and tumour

angiogenesis¹⁰¹. In pancreatic and prostate cancers, YAP and TAZ activation in tumour cells is essential for recruiting MDSCs. Abnormal YAP activity impedes the resolution of pancreatitis and support the development of pancreatic ductal adenocarcinoma in *Kras*^{G12D} and *Trp53*^{R172H} double-mutant mice, in part by promoting the accumulation of CD11b⁺/Gr1⁺ MDSCs in the damaged pancreas. Upon YAP knockout in *Kras*^{G12D} and *Trp53*^{R172H} mutant pancreatic epithelial cells, MDSC infiltration is reduced, resulting in reactivation of T lymphocytes and apoptosis of neoplastic ductal cells⁹⁸. Similarly, in a mouse model of prostate cancer (induced by combined *Pten* and *Smad4* knockout), *Yap* knockdown in tumour cells reduced MDSC recruitment and the tumour burden¹⁰². Mechanistically, YAP in complex with TEAD in cancer cells stimulates the recruitment of MDSCs within the TME by transcriptionally inducing the production of cytokines, including IL-6, macrophage colony-stimulating factor (also known as CSF1), granulocyte-macrophage colony-stimulating factor (also known as CSF2), granulocyte colony-stimulating factor (also known as CSF3) and C-X-C motif chemokine ligand 5 (CXCL5)^{98,102}. This paradigm highlights the challenges and opportunities associated with dissection of the multifaceted cell-autonomous versus non-cell-autonomous effects of YAP in tumorigenesis.

Immune-suppression. T cell activity is restricted by intrinsic immune checkpoints in order to maintain peripheral tolerance, to maintain T cell responses within physiological ranges and avoid undesired tissue damage. One of the underlying mechanisms is the interaction between programmed cell death 1 (PD1) on activated T lymphocytes and its ligand PDL1 on somatic cells. If tumour cells acquire the expression of PDL1, they can turn off

Nonalcoholic steatohepatitis

Liver pathology (not associated with alcohol abuse) characterized by lipopapoptotic hepatocyte damage, inflammation and often fibrosis, which can evolve into cirrhosis and is associated with a higher risk of developing hepatocellular carcinoma.

Myeloid-derived suppressor cells

A heterogeneous population of immature cells of myeloid origin that expand during cancer, inflammation and infection, and that have a remarkable ability to suppress T cell responses.

Immune checkpoints

Stimulatory and inhibitory pathways that regulate the type, magnitude and duration of immune responses, allowing the immune system to react to infections while protecting tissues from any harm that may derive from excessive immune responses.

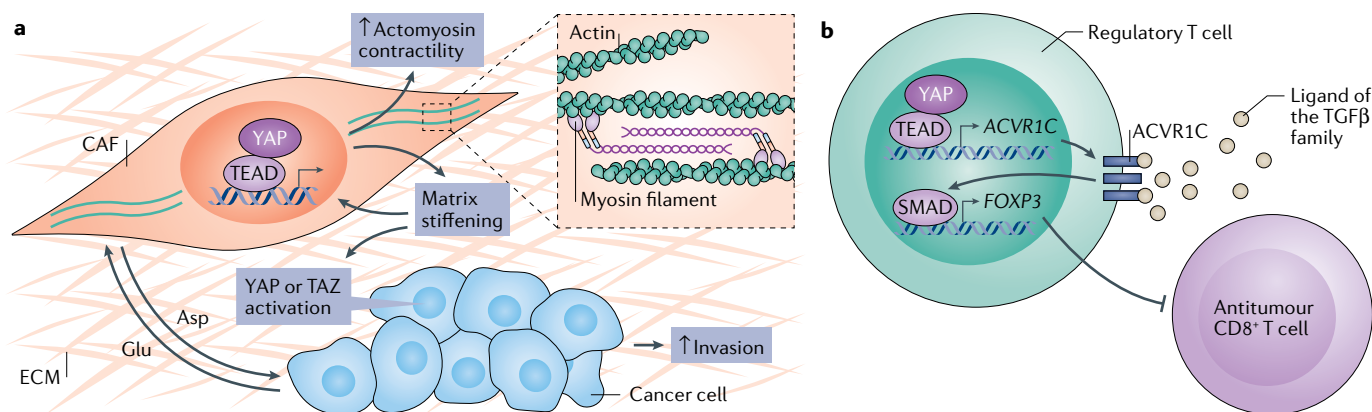


Fig. 4 | YAP and TAZ are activated in stromal cells. YAP and TAZ are activated in non-cancer cells of the tumour stroma, such as cancer-associated fibroblasts (CAFs) and regulatory T (T_{reg}) cells. **a** | Both cancer cells and CAFs actively remodel the extracellular matrix (ECM), contributing to the generation of a tumour-prone microenvironment with a stiff matrix that activates YAP and TAZ in both tumour cells and CAFs. Moreover, YAP and TAZ-dependent mechanosignalling is responsible for metabolic crosstalk between cancer cells and CAFs, whereby they exchange essential amino acids. Only YAP is shown in panel **a** for simplicity, but TAZ can carry out the same functions. **b** | YAP inside T_{reg} cells amplifies transforming growth factor- β (TGF β) signalling by upregulating the activin receptor ACVR1C, in turn leading to a boost in the expression of the T_{reg} cell-specific transcription factor FOXP3 and in FOXP3-dependent inhibition of antitumour cytotoxic T cells¹¹⁰.

tumour-specific effector T cells and escape antitumour immunity^{103,104}. Some recent reports suggest that YAP and TAZ might control the transcription of PDL1 in melanoma, lung and mesothelioma cells, and potentially other cancer cells^{105–107}. In vitro, YAP-induced PDL1 expression protects melanoma cells from T cell cytotoxicity, and treatment with PD1 inhibitors (which prevent the interaction between PD1 and its ligand) can restore T cell effector functions, leading to killing of YAP-expressing melanoma cells¹⁰⁵. These findings raise the possibility that PD1 or PDL1 inhibitors could be effective against YAP and TAZ-driven tumours. Currently, only few data are available to evaluate the relevance of this intriguing connection for human tumours, showing a trend for low PDL1 expression in YAP-low or negative tumours^{105–107}. However, a large fraction of “YAP-high” human tumours also do not express PDL1, raising the possibility that the YAP–PDL1 connection may earmark specific tumour subtypes.

YAP and TAZ in immune cells

The immune system is integral to tissue homeostasis as it can recognize incipient cancer cells and eliminate them, and as such contributes to the multilayered protection against cancer occurring at the tissue level. However, tumour cells eventually manage to ward off the attacks from the innate and adaptive arms of the immune system, and even to corrupt these friends into foes. Intriguingly, YAP and TAZ signalling within immune cells is a critical effector of these responses.

Regulatory T (T_{reg}) cells are a specialized population of T lymphocytes that suppress immune responses, maintaining self-tolerance and immune homeostasis. Once activated by a specific antigen, T_{reg} cells suppress effector T cells by secretion of immunosuppressive cytokines and depletion of essential trophic and co-stimulatory signals, inducing apoptosis or anergy of

effector T cells¹⁰⁸. It is widely recognized that T_{reg} cells play a critical role in the control of antitumour immune responses: a disproportional number of T_{reg} cells infiltrate multiple tumour types, and their relative abundance is often associated with poor clinical prognosis. Moreover, systemic removal of T_{reg} cells can elicit antitumour T cell responses¹⁰⁹. YAP1 is undetectable in isolated naive CD4⁺ T cells, but is upregulated upon T_{reg} cell skewing in vitro. Remarkably, YAP is essential in T_{reg} cells to enforce their capacity to suppress antitumour immunity (FIG. 4b). Tumour formation by mouse melanoma, colon adenocarcinoma or thymoma cells injected into syngeneic hosts was impaired upon YAP depletion specifically in T_{reg} cells. *Yap* knockout was associated with a reduced T_{reg} cell intratumoural representation, and with higher production of interferon- γ and tumour necrosis factor by CD8⁺ and CD4⁺ effector cells (markers of increased cytotoxic activity). YAP activity within T_{reg} cells is important for their accumulation and suppressive function in the TME, thus favouring the establishing of an immune-suppressive environment and the survival of neoplastic cells¹¹⁰.

Conclusions

In this Review, we have highlighted the emerging role of YAP and TAZ as a shared molecular “currency” bidirectionally exchanged by cancer cells and stromal cells to harmonize their fate and behaviour within the TME. YAP and TAZ are mechanotransducers^{73,77}, and the collective evidence suggests that the progressive corruption of the mechanical properties of the tumour ECM, with associated altered tensional homeostasis within cells, acts as the primary orchestrator of tumour-tissue biology through YAP and TAZ activation.

Here, we propose that investigating YAP and TAZ biology in stromal cells and from a tumour-tissue perspective may unveil new insights into tumour

Humanized mouse models

Immunodeficient mice engrafted with functional human haematopoietic stem cells that develop into functional human immune systems.

Mass-imaging technologies

Highly multiplexed epitope-based imaging approaches, based on immunofluorescence or mass cytometry, that enable the analysis of dozens of proteins in thousands of cells in a single experiment while preserving tissue-level information on cell position and cell–cell interactions.

organization, the enigmatic self-organizing logic of the TME and tumour–stroma co-evolution. There are several outstanding questions for future studies. To start, the difference between wound-healing and tumorigenesis is that the latter never resolves, turning tissue into their deranged caricature. Ultimately, oncogenic lesions must be responsible for such difference, but how? Can YAP and TAZ be used as a beacon to enter this issue? We have only scant understanding of how the main driving lesions may affect YAP and TAZ mechanosignalling properties of incipient tumour cells, and of how oncogenes cooperate with pre-existing or concomitant YAP and TAZ activating extracellular inputs from the surrounding stroma. Recent technological advancements allow reverse-engineering of the TME and use primary normal cells instead of transformed cell lines as the starting material, combining ex vivo genetic manipulations of tumour organoids with new biomaterials able to accommodate chemical, physical and cellular features of real tissue niches^{111–114}. In addition, the use of humanized mouse models, or tumour cell transplantation between syngeneic cohorts, should facilitate testing the role of YAP and TAZ in epithelial versus various stromal compartments in immunocompetent settings and at all stages of tumorigenesis.

The genetic dissection of YAP and TAZ in experimental tumours has so far focused on their function in epithelial cells, and exclusively in the context of tumour initiation by the main oncogenic drivers, such as RAS, WNT or SHH signalling. These investigations revealed the essentiality of YAP and TAZ for tumour emergence, which has been mostly ascribed to the cell-autonomous properties of YAP and TAZ. However, this may not be the whole story, as an equally plausible but underexplored possibility is that YAP and TAZ-depleted tumour cells may be eradicated mainly by non-cell-autonomous mechanisms, for example, cell competition or stromal cells getting normalized after weakening YAP and TAZ in cancer cells. More sophisticated mouse models are needed to enable testing the relevance of YAP and TAZ in fully established malignancies. Incidentally, a main player of the TME missing in this Review is the endothelium. Tumour-derived angiogenic factors lead to formation of new blood capillaries supporting cancer cell growth and metastasis². YAP and TAZ mechanotransduction is a relevant regulator of angiogenesis for the sprouting of blood vessels in development and in atherosclerosis^{115–119}. Surprisingly, no study has directly tackled the role of YAP and TAZ in tumour neovascularization through intratumoural vascular-specific ablation of YAP and TAZ.

Another question relates to how many vulnerabilities exist in the stroma. Are there main “hubs” in the

network that we have described, to what extent are these shared or unique to each tissue and to what extent does YAP and TAZ signalling underlie the function of these hubs? Single-cell RNA-sequencing offers insight into cellular and molecular heterogeneity of human cancer and stromal cell subtypes (as exemplified elsewhere^{120–122}) and could be employed to capture YAP and TAZ activity in each of these subtypes. A limitation here is that current single-cell analyses require tissue dissociation, leaving us blind to the spatiotemporal dynamics of cell–cell neighbourhood relationships during tumour evolution. However, this limitation is being overcome, at least in part, by novel mass-imaging technologies allowing simultaneous detection of dozens of different markers at once¹²³.

One of the most appealing new aspects of YAP and TAZ biology outlined here is the role of YAP and TAZ in immune cells. The clinical success of immune-checkpoint inhibitors opened a new era in cancer medicine, in fact, highlighting the power of tackling tumour–stroma interactions. Unfortunately, however, typically only a fraction of patients responds to these new drugs^{103,104}. The requirement of YAP in T_{reg} cells for tumorigenesis¹¹⁰ at least suggests that combining immune-checkpoint inhibitors with YAP and TAZ inhibitors, such as BET inhibitors, might be plausible to induce more robust responses in a wider number of patients. The findings also invite a more in-depth investigation into the role of YAP and TAZ in T_{reg} cells in contexts other than cancer, and more generally in immune cells called into action by special tissue demands, such as in the context of tissue repair, fibrosis or autoimmune disorders. Are YAP and TAZ relevant also in these conditions? If so, in what cell subtype? Of note, and potentially in line with this view, YAP has been shown to be crucial in the epicardium of infarcted hearts to locally recruit T cells¹²⁴.

To conclude, current “personalized medicine” approaches target cancer’s diseased genes, which undergo continuous selection and evolution — perhaps explaining the partial success of this approach. However, tumours are also tissues with their own developmental constraints, where organization of one component is critical to sustain the next component and so forth. The discovery of YAP and TAZ at the centrepiece of the intricate networks of signals exchanged within the TME indicates that identification of their effectors may lead to the identification of prognostic markers and new targets for therapy. By reverse-engineering the lessons learned from YAP and TAZ functions at the tumour-tissue level, new insights will emerge into how to prevent tumorigenesis and metastasis, and even diminish established lesions.

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1. Bissell, M. J. & Radisky, D. Putting tumours in context. *Nat. Rev. Cancer* **1**, 46–54 (2001).
2. Egeblad, M., Nakasone, E. S. & Werb, Z. Tumors as organs: complex tissues that interface with the entire organism. *Dev. Cell* **18**, 884–901 (2010).
3. Dvorak, H. F. Tumors: wounds that do not heal. Similarities between tumor stroma generation and wound healing. *N. Engl. J. Med.* **315**, 1650–1659 (1986).
4. Bissell, M. J. & Hines, W. C. Why don’t we get more cancer? A proposed role of the microenvironment in

- restraining cancer progression. *Nat. Med.* **17**, 320–329 (2011).
5. Quail, D. F. & Joyce, J. A. Microenvironmental regulation of tumor progression and metastasis. *Nat. Med.* **19**, 1423–1437 (2013).
6. Martincorena, I. et al. Tumor evolution. High burden and pervasive positive selection of somatic mutations in normal human skin. *Science* **348**, 880–886 (2015).
7. Yokoyama, A. et al. Age-related remodelling of oesophageal epithelia by mutated cancer drivers. *Nature* **565**, 312–317 (2019).

This paper demonstrates that clones of cells bearing mutations in driver oncogenes emerge multifocally and expand with ageing in the normal oesophageal epithelium of healthy people.

8. Battle, E. & Clevers, H. Cancer stem cells revisited. *Nat. Med.* **23**, 1124–1134 (2017).
9. Overholtzer, M. et al. Transforming properties of YAP, a candidate oncogene on the chromosome 11q22 amplicon. *Proc. Natl Acad. Sci. USA* **103**, 12405–12410 (2006).

10. Zender, L. et al. Identification and validation of oncogenes in liver cancer using an integrative oncogenomic approach. *Cell* **125**, 1253–1267 (2006).
11. Chan, S. W. et al. A role for TAZ in migration, invasion, and tumorigenesis of breast cancer cells. *Cancer Res.* **68**, 2592–2598 (2008).
12. Piccolo, S., Dupont, S. & Cordenonsi, M. The biology of YAP/TAZ: hippo signaling and beyond. *Physiol. Rev.* **94**, 1287–1312 (2014).
13. Hansen, C. G., Moroishi, T. & Guan, K. L. YAP and TAZ: a nexus for Hippo signaling and beyond. *Trends Cell Biol.* **25**, 499–513 (2015).
14. Moroishi, T., Hansen, C. G. & Guan, K. L. The emerging roles of YAP and TAZ in cancer. *Nat. Rev. Cancer* **15**, 73–79 (2015).
15. Zanconato, F., Cordenonsi, M. & Piccolo, S. YAP/TAZ at the roots of cancer. *Cancer Cell* **29**, 783–803 (2016).
16. Wang, Y. et al. Comprehensive molecular characterization of the Hippo signaling pathway in cancer. *Cell Rep.* **25**, 1304–1317 (2018).
17. Moya, I. M. & Halder, G. Hippo-YAP/TAZ signalling in organ regeneration and regenerative medicine. *Nat. Rev. Mol. Cell Biol.* **20**, 211–226 (2019).
18. Molyneux, G. et al. BRCA1 basal-like breast cancers originate from luminal epithelial progenitors and not from basal stem cells. *Cell Stem Cell* **7**, 403–417 (2010).
19. Xu, X. et al. Evidence for type II cells as cells of origin of K-Ras-induced distal lung adenocarcinoma. *Proc. Natl Acad. Sci. USA* **109**, 4910–4915 (2012).
20. Friedmann-Morvinski, D. et al. Dedifferentiation of neurons and astrocytes by oncogenes can induce gliomas in mice. *Science* **338**, 1080–1084 (2012).
21. Kopp, J. L. et al. Identification of Sox9-dependent acinar-to-ductal reprogramming as the principal mechanism for initiation of pancreatic ductal adenocarcinoma. *Cancer Cell* **22**, 737–750 (2012).
22. Sia, D., Villanueva, A., Friedman, S. L. & Llovet, J. M. Liver cancer cell of origin, molecular class, and effects on patient prognosis. *Gastroenterology* **152**, 745–761 (2017).
23. Panciera, T. et al. Induction of expandable tissue-specific stem/progenitor cells through transient expression of YAP/TAZ. *Cell Stem Cell* **19**, 725–737 (2016).
24. Zanconato, F. et al. Genome-wide association between YAP/TAZ/TEAD and AP-1 at enhancers drives oncogenic growth. *Nat. Cell Biol.* **17**, 1218–1227 (2015).
25. Kapoor, A. et al. Yap1 activation enables bypass of oncogenic Kras addition in pancreatic cancer. *Cell* **158**, 185–197 (2014).
26. Jang, W., Kim, T., Koo, J. S., Kim, S. K. & Lim, D. S. Mechanical cue-induced YAP instructs Skp2-dependent cell cycle exit and oncogenic signaling. *EMBO J.* **36**, 2510–2528 (2017).
27. Gill, M. K. et al. A feed forward loop enforces YAP/TAZ signaling during tumorigenesis. *Nat. Commun.* **9**, 3510 (2018).
28. Yuan, W. C. et al. NUA2 is a critical YAP target in liver cancer. *Nat. Commun.* **9**, 4834 (2018).
29. Maglic, D. et al. YAP-TEAD signaling promotes basal cell carcinoma development via a c-JUN/AP1 axis. *EMBO J.* **37**, e98642 (2018).
30. Liu, X. et al. Tead and AP1 coordinate transcription and motility. *Cell Rep.* **14**, 1169–1180 (2016).
31. Croci, O. et al. Transcriptional integration of mitogenic and mechanical signals by Myc and YAP. *Genes Dev.* **31**, 2017–2022 (2017).
32. Choi, W. et al. YAP/TAZ initiates gastric tumorigenesis via upregulation of MYC. *Cancer Res.* **78**, 3306–3320 (2018).
33. Koo, J. H. & Guan, K. L. Interplay between YAP/TAZ and metabolism. *Cell Metab.* **28**, 196–206 (2018).
34. Cosset, E. et al. Glut3 addition is a druggable vulnerability for a molecularly defined subpopulation of glioblastoma. *Cancer Cell* **32**, 856–868 (2017).
35. Cox, A. G. et al. Yap reprograms glutamine metabolism to increase nucleotide biosynthesis and enable liver growth. *Nat. Cell Biol.* **18**, 886–896 (2016).
36. Santion, G. et al. dNTP metabolism links mechanical cues and YAP/TAZ to cell growth and oncogene-induced senescence. *EMBO J.* **37**, e97780 (2018).
37. Yilmam, D. et al. Hippo pathway activity influences liver cell fate. *Cell* **157**, 1324–1338 (2014).
38. Yui, S. et al. YAP/TAZ-dependent reprogramming of colonic epithelium links ECM remodeling to tissue regeneration. *Cell Stem Cell* **22**, 35–49 (2018).
39. Ayyaz, A. et al. Single-cell transcriptomes of the regenerating intestine reveal a revival stem cell. *Nature* **569**, 121–125 (2019).
40. Serra, D. et al. Self-organization and symmetry breaking in intestinal organoid development. *Nature* **569**, 66–72 (2019).
41. Cordenonsi, M. et al. The Hippo transducer TAZ confers cancer stem cell-related traits on breast cancer cells. *Cell* **147**, 759–772 (2011).
42. Mani, S. A. et al. The epithelial–mesenchymal transition generates cells with properties of stem cells. *Cell* **133**, 704–715 (2008).
43. Tang, Y., Feinberg, T., Keller, E. T., Li, X. Y. & Weiss, S. J. Snail/Slug binding interactions with YAP/TAZ control skeletal stem cell self-renewal and differentiation. *Nat. Cell Biol.* **18**, 917–929 (2016).
44. Zhao, B. et al. TEAD mediates YAP-dependent gene induction and growth control. *Genes Dev.* **22**, 1962–1971 (2008).
45. Chang, L. et al. The SWI/SNF complex is a mechanoregulated inhibitor of YAP and TAZ. *Nature* **563**, 265–269 (2018).
46. Zanconato, F., Battilana, G., Cordenonsi, M. & Piccolo, S. YAP/TAZ as therapeutic targets in cancer. *Curr. Opin. Pharmacol.* **29**, 26–33 (2016).
47. Kim, M. H. & Kim, J. Role of YAP/TAZ transcriptional regulators in resistance to anti-cancer therapies. *Cell. Mol. Life Sci.* **74**, 1457–1474 (2017).
48. Lin, L. et al. The Hippo effector YAP promotes resistance to RAF- and MEK-targeted cancer therapies. *Nat. Genet.* **47**, 250–256 (2015).
49. Kim, M. H. et al. Actin remodeling confers BRAF inhibitor resistance to melanoma cells through YAP/TAZ activation. *EMBO J.* **35**, 462–478 (2016).
50. Shao, D. D. et al. KRAS and YAP1 converge to regulate EMT and tumor survival. *Cell* **158**, 171–184 (2014).
51. Li, Z. et al. Loss of the FAT1 tumor suppressor promotes resistance to CDK4/6 inhibitors via the Hippo pathway. *Cancer Cell* **34**, 893–905 (2018).
52. Lin, C. H. et al. Microenvironment rigidity modulates responses to the HER2 receptor tyrosine kinase inhibitor lapatinib via YAP and TAZ transcription factors. *Mol. Biol. Cell* **26**, 3946–3953 (2015).
53. Zanconato, F. et al. Transcriptional addition in cancer cells is mediated by YAP/TAZ through BRD4. *Nat. Med.* **24**, 1599–1610 (2018).
54. Warren, J. S. A., Xiao, Y. & Lamar, J. M. YAP/TAZ activation as a target for treating metastatic cancer. *Cancers* **10**, E115 (2018).
55. Qiao, Y. et al. YAP regulates actin dynamics through ARHGAP29 and promotes metastasis. *Cell Rep.* **19**, 1495–1502 (2017).
56. Mason, D. E. et al. YAP and TAZ limit cytoskeletal and focal adhesion maturation to enable persistent cell motility. *J. Cell Biol.* **218**, 1369–1389 (2019).
57. Zhao, B. et al. Cell detachment activates the Hippo pathway via cytoskeleton reorganization to induce anoikis. *Genes Dev.* **26**, 54–68 (2012).
58. Haemmerle, M. et al. Platelets reduce anoikis and promote metastasis by activating YAP1 signaling. *Nat. Commun.* **8**, 310 (2017).
59. Lee, C. K. et al. Tumor metastasis to lymph nodes requires YAP-dependent metabolic adaptation. *Science* **363**, 644–649 (2019).
60. Amoyel, M. & Bach, E. A. Cell competition: how to eliminate your neighbours. *Development* **141**, 988–1000 (2014).
61. Di Gregorio, A., Bowling, S. & Rodriguez, T. A. Cell competition and its role in the regulation of cell fitness from development to cancer. *Dev. Cell* **38**, 621–634 (2016).
62. Chen, C. L., Schroeder, M. C., Kango-Singh, M., Tao, C. & Halder, G. Tumor suppression by cell competition through regulation of the Hippo pathway. *Proc. Natl Acad. Sci. USA* **109**, 484–489 (2012).
63. Norman, M. et al. Loss of Scribble causes cell competition in mammalian cells. *J. Cell Sci.* **125**, 59–66 (2012).
64. Lin, W. H., Asmann, Y. W. & Anastasiadis, P. Z. Expression of polarity genes in human cancer. *Cancer Inform.* **14**, 15–28 (2015).
65. Grzeschik, N. A., Parsons, L. M., Allott, M. L., Harvey, K. F. & Richardson, H. E. Lgl, aPKC, and Crumbs regulate the Salvador/Warts/Hippo pathway through two distinct mechanisms. *Curr. Biol.* **20**, 573–581 (2010).
66. Wagstaff, L. et al. Mechanical cell competition kills cells via induction of lethal p53 levels. *Nat. Commun.* **7**, 11373 (2016).
67. Aragone, M. et al. A mechanical checkpoint controls multicellular growth through YAP/TAZ regulation by actin-processing factors. *Cell* **154**, 1047–1059 (2013).
68. Liu, Z. et al. Differential YAP expression in glioma cells induces cell competition and promotes tumorigenesis. *J. Cell Sci.* **132**, jcs225714 (2019).
69. Azzolin, L. et al. YAP/TAZ incorporation in the β -catenin destruction complex orchestrates the Wnt response. *Cell* **158**, 157–170 (2014).
70. Gregoroff, A., Liu, Y., Inanlou, M. R., Khomchuk, Y. & Wana, J. L. YAP-dependent reprogramming of Lgr5⁺ stem cells drives intestinal regeneration and cancer. *Nature* **526**, 715–718 (2015).
71. Cai, J., Maitra, A., Anders, R. A., Taketo, M. M. & Pan, D. β -Catenin destruction complex-independent regulation of Hippo-YAP signaling by APC in intestinal tumorigenesis. *Genes Dev.* **29**, 1493–1506 (2015).
72. Suijkerbuijk, S. J., Kolahgar, G., Kucinski, I. & Piddini, E. Cell competition drives the growth of intestinal adenomas in *Drosophila*. *Curr. Biol.* **26**, 428–438 (2016).
73. Cai, J., Maitra, A., Anders, R. A., Taketo, M. M. & Pan, D. β -Catenin destruction complex-independent regulation of Hippo-YAP signaling by APC in intestinal tumorigenesis. *Genes Dev.* **29**, 1493–1506 (2015).
74. Dotto, G. P. Multifocal epithelial tumors and field cancerization: stroma as a primary determinant. *J. Clin. Invest.* **124**, 1446–1453 (2014).
75. Brusatin, G., Panciera, T., Gandini, A., Citron, A. & Piccolo, S. Biomaterials and engineered microenvironments to control YAP/TAZ-dependent cell behaviour. *Nat. Mater.* **17**, 1063–1075 (2018).
76. Northey, J. J., Przybyla, L. & Weaver, V. M. Tissue force programs cell fate and tumor aggression. *Cancer Discov.* **7**, 1224–1237 (2017).
77. Dupont, S. et al. Role of YAP/TAZ in mechanotransduction. *Nature* **474**, 179–183 (2011).
78. Provenzano, P. P. & Keely, P. J. Mechanical signaling through the cytoskeleton regulates cell proliferation by coordinated focal adhesion and Rho GTPase signaling. *J. Cell Sci.* **124**, 1195–1205 (2011).
79. Stein, C. et al. YAP1 exerts its transcriptional control via TEAD-mediated activation of enhancers. *PLOS Genet.* **11**, e1005465 (2015).
80. Chang, C. et al. A laminin 511 matrix is regulated by TAZ and functions as the ligand for the α 6 β 1 integrin to sustain breast cancer stem cells. *Genes Dev.* **29**, 1–6 (2015).
81. Lee, J. et al. Tissue transglutaminase mediated tumor–stroma interaction promotes pancreatic cancer progression. *Clin. Cancer Res.* **21**, 4482–4493 (2015).
82. Katsumi, A., Naoe, T., Matsushita, T., Kaibuchi, K. & Schwartz, M. A. Integrin activation and matrix binding mediate cellular responses to mechanical stretch. *J. Biol. Chem.* **280**, 16546–16549 (2005).
83. Nardone, G. et al. YAP regulates cell mechanics by controlling focal adhesion assembly. *Nat. Commun.* **8**, 15321 (2017).
84. Calvo, F. et al. Mechanotransduction and YAP-dependent matrix remodelling is required for the generation and maintenance of cancer-associated fibroblasts. *Nat. Cell Biol.* **15**, 637–646 (2013).
85. Ferrari, N. et al. Dickkopf-3 links HSF1 and YAP/TAZ signalling to control aggressive behaviours in cancer-associated fibroblasts. *Nat. Commun.* **10**, 130 (2019).
86. Bertero, T. et al. Tumor–stroma mechanics coordinate amino acid availability to sustain tumor growth and malignancy. *Cell Metab.* **29**, 124–140 (2019).

87. Zhang, W. et al. Downstream of mutant KRAS, the transcription regulator YAP is essential for neoplastic progression to pancreatic ductal adenocarcinoma. *Sci. Signal.* **7**, ra42 (2014).
88. Er, E. E. et al. Pericyte-like spreading by disseminated cancer cells activates YAP and MRTF for metastatic colonization. *Nat. Cell Biol.* **20**, 966–978 (2018).
89. Albregues, J. et al. Neutrophil extracellular traps produced during inflammation awaken dormant cancer cells in mice. *Science* **361**, eaao4227 (2018).
90. Pierce, J. P. et al. Lifetime cigarette smoking and breast cancer prognosis in the After Breast Cancer Pooling Project. *J. Natl Cancer Inst.* **106**, djt359 (2014).
91. Taniguchi, K. et al. A gp130-Src-YAP module links inflammation to epithelial regeneration. *Nature* **519**, 57–62 (2015).
92. Long, K. B., Collier, A. I. & Beatty, G. L. Macrophages: key orchestrators of a tumor microenvironment defined by therapeutic resistance. *Mol. Immunol.* **110**, 3–12 (2017).
93. Lu, L. et al. Hippo signaling is a potent in vivo growth and tumor suppressor pathway in the mammalian liver. *Proc. Natl Acad. Sci. USA* **107**, 1437–1442 (2010).
94. He, G. B. et al. Identification of liver cancer progenitors whose malignant progression depends on autocrine IL-6 signaling. *Cell* **155**, 384–396 (2013).
95. Guo, X. et al. Single tumor-initiating cells evade immune clearance by recruiting type II macrophages. *Genes Dev.* **31**, 247–259 (2017).
- This work shows that activation of YAP in a single hepatocyte is sufficient to recruit macrophages, which help the initiating tumour escape immune surveillance.**
96. Kim, W. et al. Hepatic Hippo signaling inhibits protumoural microenvironment to suppress hepatocellular carcinoma. *Gut* **67**, 1692–1703 (2018).
97. Wang, X. et al. Hepatocyte TAZ/WWTR1 promotes inflammation and fibrosis in nonalcoholic steatohepatitis. *Cell Metab.* **24**, 848–862 (2016).
98. Murakami, S. et al. Yes-associated protein mediates immune reprogramming in pancreatic ductal adenocarcinoma. *Oncogene* **36**, 1232–1244 (2017).
99. McWhorter, F. Y., Wang, T., Nguyen, P., Chung, T. & Liu, W. F. Modulation of macrophage phenotype by cell shape. *Proc. Natl Acad. Sci. USA* **110**, 17253–17258 (2013).
100. McWhorter, F. Y., Davis, C. T. & Liu, W. F. Physical and mechanical regulation of macrophage phenotype and function. *Cell. Mol. Life Sci.* **72**, 1303–1316 (2015).
101. Gabrilovich, D. I. Myeloid-derived suppressor cells. *Cancer Immunol. Res.* **5**, 3–8 (2017).
102. Wang, G. et al. Targeting YAP-dependent MDSC infiltration impairs tumor progression. *Cancer Discov.* **6**, 80–95 (2016).
- This work shows that MDSCs are instrumental for prostate tumour progression, and that their recruitment is mediated by YAP activation in cancer cells.**
103. Wei, S. C., Duffy, C. R. & Allison, J. P. Fundamental mechanisms of immune checkpoint blockade therapy. *Cancer Discov.* **8**, 1069–1086 (2018).
104. Ribas, A. & Wolchok, J. D. Cancer immunotherapy using checkpoint blockade. *Science* **359**, 1350–1355 (2018).
105. Kim, M. H. et al. YAP-induced PD-L1 expression drives immune evasion in BRAFi-resistant melanoma. *Cancer Immunol. Res.* <https://doi.org/10.1158/2326-6066.CIR-17-0320> (2018).
106. Miao, J. et al. YAP regulates PD-L1 expression in human NSCLC cells. *Oncotarget* **8**, 114576–114587 (2017).
107. Hsu, P. C. et al. Inhibition of yes-associated protein down-regulates PD-L1 (CD274) expression in human malignant pleural mesothelioma. *J. Cell. Mol. Med.* **22**, 3139–3148 (2018).
108. Nishikawa, H. & Sakaguchi, S. Regulatory T cells in tumor immunity. *Int. J. Cancer* **127**, 759–767 (2010).
109. Tanaka, A. & Sakaguchi, S. Regulatory T cells in cancer immunotherapy. *Cell Res.* **27**, 109–118 (2017).
110. Ni, X. et al. YAP is essential for Treg-mediated suppression of antitumor immunity. *Cancer Discov.* **8**, 1026–1043 (2018).
- This work is the first demonstration that genetic ablation of YAP in Treg cells impairs their capacity to suppress antitumour immunity in a mouse model of melanoma.**
111. Neal, J. T. et al. Organoid modeling of the tumor immune microenvironment. *Cell* **175**, 1972–1988 (2018).
112. Drost, J. & Clevers, H. Organoids in cancer research. *Nat. Rev. Cancer* **18**, 407–418 (2018).
113. Gjorevski, N. et al. Designer matrices for intestinal stem cell and organoid culture. *Nature* **539**, 560–564 (2016).
- In this work the authors developed synthetic mechanically dynamic matrices for the expansion of intestinal stem cells, allowing one to gain insight into how the 3D microenvironment influences intestinal stem cell proliferation, fate decisions and self-organization, and the role of YAP in these processes.**
114. Balani, S., Nguyen, L. V. & Eaves, C. J. Modeling the process of human tumorigenesis. *Nat. Commun.* **8**, 15422 (2017).
115. Wang, X. et al. YAP/TAZ orchestrate VEGF signaling during developmental angiogenesis. *Dev. Cell* **42**, 462–478 (2017).
116. Kim, J. et al. YAP/TAZ regulates sprouting angiogenesis and vascular barrier maturation. *J. Clin. Invest.* **127**, 3441–3461 (2017).
117. Nakajima, H. et al. Flow-dependent endothelial YAP regulation contributes to vessel maintenance. *Dev. Cell* **40**, 523–536 (2017).
118. Wang, L. et al. Integrin-YAP/TAZ-JNK cascade mediates atheroprotective effect of unidirectional shear flow. *Nature* **540**, 579–582 (2016).
119. Wang, K. C. et al. Flow-dependent YAP/TAZ activities regulate endothelial phenotypes and atherosclerosis. *Proc. Natl Acad. Sci. USA* **113**, 11525–11530 (2016).
120. Tirosh, I. et al. Dissecting the multicellular ecosystem of metastatic melanoma by single-cell RNA-seq. *Science* **352**, 189–196 (2016).
121. Kumar, M. P. et al. Analysis of single-cell RNA-Seq identifies cell–cell communication associated with tumor characteristics. *Cell Rep.* **25**, 1458–1468 (2018).
122. Puram, S. V. et al. Single-cell transcriptomic analysis of primary and metastatic tumor ecosystems in head and neck cancer. *Cell* **171**, 1611–1624 (2017).
123. Bodenmiller, B. Multiplexed epitope-based tissue imaging for discovery and healthcare applications. *Cell Syst.* **2**, 225–238 (2016).
124. Ramjee, V. et al. Epicardial YAP/TAZ orchestrate an immunosuppressive response following myocardial infarction. *J. Clin. Invest.* **127**, 899–911 (2017).
125. Totaro, A., Panciera, T. & Piccolo, S. YAP/TAZ upstream signals and downstream responses. *Nat. Cell Biol.* **20**, 888–899 (2018).
126. Low, B. C. et al. YAP/TAZ as mechanosensors and mechanotransducers in regulating organ size and tumor growth. *FEBS Lett.* **588**, 2663–2670 (2014).
127. Meng, Z., Moroishi, T. & Guan, K. L. Mechanisms of Hippo pathway regulation. *Genes Dev.* **30**, 1–17 (2016).
128. Totaro, A., Castellan, M., Di Biagio, D. & Piccolo, S. Crosstalk between YAP/TAZ and Notch signaling. *Trends Cell Biol.* **28**, 560–573 (2018).
129. Ege, N. et al. Quantitative analysis reveals that actin and Src-family kinases regulate nuclear YAP1 and its export. *Cell Syst.* **6**, 692–708 (2018).
130. Galli, C. G. et al. YAP drives growth by controlling transcriptional pause release from dynamic enhancers. *Mol. Cell* **60**, 328–337 (2015).
131. Monroe, T. O. et al. YAP partially reprograms chromatin accessibility to directly induce adult cardiogenesis in vivo. *Dev. Cell* **48**, 765–779 (2019).
132. Diamantopoulou, Z. et al. TIAM1 antagonizes TAZ/YAP both in the destruction complex in the cytoplasm and in the nucleus to inhibit invasion of intestinal epithelial cells. *Cancer Cell* **31**, 621–634 (2017).
133. Kim, J. et al. Long noncoding RNA MALAT1 suppresses breast cancer metastasis. *Nat. Genet.* **50**, 1705–1715 (2018).
134. Zhang, W. et al. VGLL4 functions as a new tumor suppressor in lung cancer by negatively regulating the YAP-TEAD transcriptional complex. *Cell Res.* **24**, 331–343 (2014).
135. Jiao, S. et al. A peptide mimicking VGLL4 function acts as a YAP antagonist therapy against gastric cancer. *Cancer Cell* **25**, 166–180 (2014).
136. Bartucci, M. et al. TAZ is required for metastatic activity and chemoresistance of breast cancer stem cells. *Oncogene* **34**, 681–690 (2015).
137. Chen, Q. et al. A temporal requirement for Hippo signaling in mammary gland differentiation, growth, and tumorigenesis. *Genes Dev.* **28**, 432–437 (2014).
138. Lo Sardo, F., Strano, S. & Blandino, G. YAP and TAZ in lung cancer: oncogenic role and clinical targeting. *Cancers* **10**, 137 (2018).
139. Zhang, W. et al. YAP promotes malignant progression of Lkb1-deficient lung adenocarcinoma through downstream regulation of survivin. *Cancer Res.* **75**, 4450–4457 (2015).
140. Patel, S. H., Camargo, F. D. & Yimlamai, D. Hippo signaling in the liver regulates organ size, cell fate, and carcinogenesis. *Gastroenterology* **152**, 533–545 (2017).
141. Rozenfurt, E., Sinnott-Smith, J. & Eibl, G. Yes-associated protein (YAP) in pancreatic cancer: at the epicenter of a targetable signaling network associated with patient survival. *Signal Transduct. Target. Ther.* **3**, 11 (2018).
142. Debaugnes, M. et al. YAP and TAZ are essential for basal and squamous cell carcinoma initiation. *EMBO Rep.* **19**, e45809 (2018).

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